Scenario:

You're asked to develop a .NET backend application that allows the management of foreign exchange rates.

- Create a ASP.NET Core Web API to perform CRUD operations of foreign exchange rates (consider that each currency pair (e.g., USD/EUR) has 2 prices [bid and ask]).
- For the endpoint that will be used to retrieve the rates for a specific currency pair, design a mechanism to check if the requested rate is available in the database.
    - If not available, fetch data from a third-party API (we suggest you use the CURRENCY_EXCHANGE_RATE function of https://www.alphavantage.co) and store it in the database.
    - If data is available in the database, return it directly.

Bonus (Optional):

- Every time a new rate is added to the database, an event should be raised using a message queue system.
- You can store the data in-memory, although, the usage of an actual database would be valued.

Submission Guidelines:

- Share your code repository (e.g., GitHub or Bitbucket) containing the source code of your application.
- Provide well-structured, clean, and commented code.
- Include documentation explaining how to set up and run the application.
- Demonstrate proper error handling and logging techniques.
- Ensure that the submitted code is properly tested, we would like to see some unit tests.
- Highlight any limitations the provided solution might have and possible improvements.